Academic Programs

B.Tech. ? Automobile Engineering

Programme Educational Objectives (PEOs)

PEO-1: Provide a strong foundation in mathematical, scientific and engineering fundamentals that enable the students to formulate, analyze and solve engineering problems and to prepare them for graduate studies.

PEO-2: Apply knowledge and concepts of automotive technology to synthesize data and solve multi-disciplinary engineering problems.

PEO-3: Work as part of teams for successful career in automotive and ancillary industry that meet the needs of Indian and multinational companies.

PEO-4: Undertake research and development projects with multi-disciplinary approach which are cost effective and efficient so as to resolve automotive engineering issues of social relevance.

PEO-5: Demonstrate their professional, ethical and social responsibilities for a successful professional career and contribute their part for addressing various global issues

Program Outcomes (POs)

Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Modern tool usage: Create, select, and
apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

The engineer and society: Apply reasoning
informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Environment and sustainability: Understand
the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

Ethics: Apply ethical principles and commit
to professional ethics and responsibilities and norms of the engineering practice.

Individual and teamwork: Function
effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Communication: Communicate effectively on
complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Project management and finance: Demonstrate
knowledge and understanding of the engineering and management principles and apply these to one?s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Life-long learning: Recognize the need for
and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO-1: Use automobile engineering
fundamentals, techniques, skills, latest modeling/design/analysis/simulation tools, software and equipment necessary to evaluate and analyze the systems in automotive design environments.

PSO-2: Address specific problems in the
field of automotive engineering in the form of projects for interpretation of data and synthesis of information to

provide valid conclusions